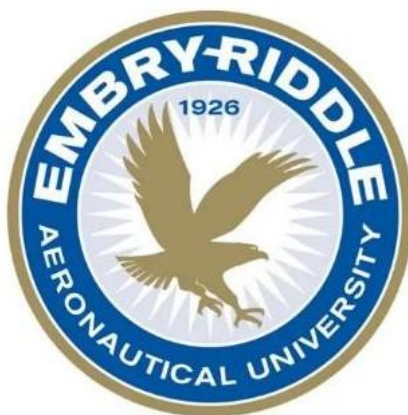


Gyro Modeling and Control



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Procedure

A. Main goal of the experiment

The main goal of this experiment is to study the behavior of a rotary gyroscope and understand how feedback control improves its performance. This includes developing and validating a mathematical model, implementing a PI controller to track a desired deflection angle α , and analyzing how the system responds under both open-loop and closed-loop conditions. The experiment also aims to examine the relationship between base rotation, gyroscopic torque, and deflection angle.

B. Experiment procedure

The experiment was carried out in multiple stages. First, the open-loop response of the system was obtained by applying a step voltage to the motor while the base was rotating. The resulting signals V_m , θ , and α were recorded to determine system characteristics such as steady-state value and time constant, and to compare with the theoretical transfer function.

Next, a PI controller was implemented in the Simulink model using the designed gains k_p and k_i . The system was then commanded to track a reference deflection angle of $\alpha = 2^\circ$. The motor voltage was adjusted automatically by the controller, and the system response was recorded to evaluate tracking performance.

Finally, disturbance tests were performed by switching the controller on and off while the base was rotated. The responses of V_m , θ , and α were observed to compare open-loop and closed-loop behavior in terms of stability, tracking accuracy, and disturbance rejection.

Results

A. Derivations and Calculations

Derivation of Gyro PI Controller

Starting from the PI control law,

$$V_m(s) = \left(k_p + \frac{k_i}{s}\right) e(s) \quad (i)$$

where

$$e(s) = \alpha_{ref}(s) - \alpha(s) \quad (ii)$$

Substituting the error expression gives

$$V_m(s) = \left(k_p + \frac{k_i}{s}\right) (\alpha_{ref}(s) - \alpha(s)) \quad (iii)$$

The open-loop gyroscope model is

$$\frac{\alpha(s)}{V_m(s)} = \frac{1}{G_g} \frac{K}{\tau s + 1} \quad (iv)$$

so

$$\alpha(s) = \frac{1}{G_g} \frac{K}{\tau s + 1} V_m(s) \quad (v)$$

Substituting $V_m(s)$ into the above expression,

$$\alpha(s) = \frac{1}{G_g} \frac{K}{\tau s + 1} \left(k_p + \frac{k_i}{s}\right) (\alpha_{ref}(s) - \alpha(s)) \quad (vi)$$

$$\alpha(s) = \frac{K(sk_p + k_i)}{G_g s(\tau s + 1)} (\alpha_{ref}(s) - \alpha(s)) \quad (vii)$$

Rearranging,

$$\alpha(s) \left[1 + \frac{K(sk_p + k_i)}{G_g s(\tau s + 1)}\right] = \frac{K(sk_p + k_i)}{G_g s(\tau s + 1)} \alpha_{ref}(s) \quad (viii)$$

Therefore,

$$\frac{\alpha(s)}{\alpha_{ref}(s)} = \frac{\frac{K(sk_p + k_i)}{G_g s(\tau s + 1)}}{1 + \frac{K(sk_p + k_i)}{G_g s(\tau s + 1)}} \quad (ix)$$

$$\frac{\alpha(s)}{\alpha_{ref}(s)} = \frac{K(sk_p + k_i)}{G_g s(\tau s + 1) + K(sk_p + k_i)} \quad (x)$$

Expanding the denominator,

$$G_{CL}(s) = \frac{\alpha(s)}{\alpha_{ref}(s)} = \frac{K(sk_p + k_i)}{G_g \tau s^2 + (Kk_p + G_g)s + Kk_i} \quad (xi)$$

PI control gains k_p and k_i in terms of ω_n and ζ

The closed-loop transfer function denominator is

$$G_g \tau s^2 + (K k_p + G_g) s + K k_i \quad (xii)$$

Dividing by $G_g \tau$,

$$s^2 + \frac{K k_p + G_g}{G_g \tau} s + \frac{K k_i}{G_g \tau} \quad (xiii)$$

Comparing with the standard second-order form

$$s^2 + 2\zeta \omega_n s + \omega_n^2 \quad (xiv)$$

gives

$$\frac{K k_p + G_g}{G_g \tau} = 2\zeta \omega_n \quad (xv)$$

$$\frac{K k_i}{G_g \tau} = \omega_n^2 \quad (xvi)$$

Solving for the gains,

$$k_p = \frac{G_g (2\zeta \omega_n \tau - 1)}{K} \quad (xvii)$$

$$k_i = \frac{G_g \tau \omega_n^2}{K} \quad (xviii)$$

Numerical PI controller design

The proportional gain is

$$k_p = \frac{G_g (2\zeta \omega_n \tau - 1)}{K} \quad (xix)$$

$$k_p = \frac{5.20(2(1.05)(3.61)(0.9) - 1)}{1.53} \quad (xx)$$

$$k_p = 19.79 \quad (xxi)$$

The integral gain is

$$k_i = \frac{G_g \tau \omega_n^2}{K} \quad (xxii)$$

$$k_i = \frac{5.20(0.9)(3.61)^2}{1.53} \quad (xxiii)$$

$$k_i = 39.86 \quad (xxiv)$$

B. Time Constant from Open-Loop Response

The time constant is obtained from the open-loop deflection response $\alpha(t)$ using the standard first-order system relation.

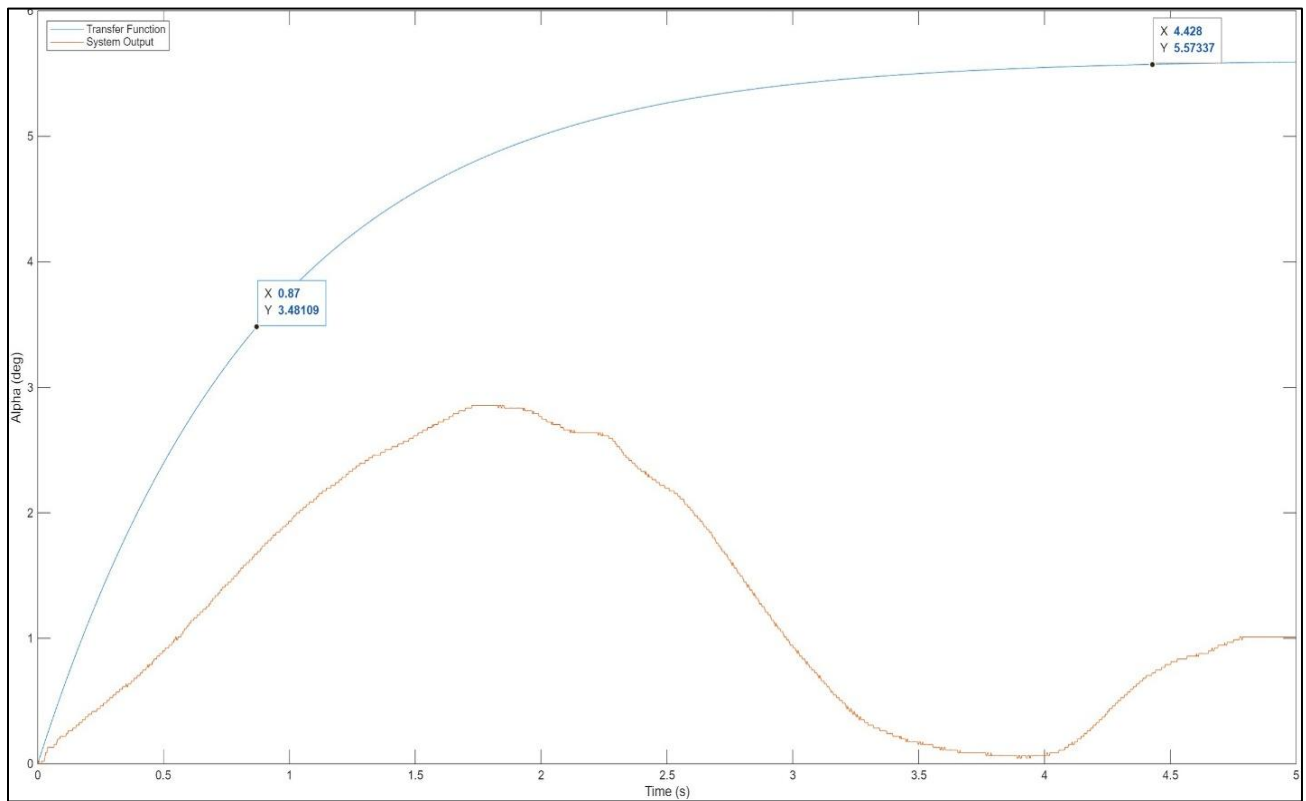


Figure (i): Open Loop – Alpha Plot

From figure (i), $\alpha_{ss} \approx 5.7^\circ$.

For a first-order system,

$$\alpha(\tau) = 0.632 \alpha_{ss} \quad (xxv)$$

$$\alpha(\tau) = 0.632 \times 5.7 \approx 3.48^\circ \quad (xxvi)$$

From figure (i), the response begins at $t_0 = 0$ s while $\alpha = 3.48^\circ$ occurs at

$$t_{63\%} = 0.87 \text{ s} \quad (xxvii)$$

Therefore,

$$\tau = t_{63\%} - t_0 = 0.87 - 0 = 0.87 \text{ s} \quad (xxviii)$$

This represents the experimental time constant of the open-loop system.

C. Open Loop Response Plots

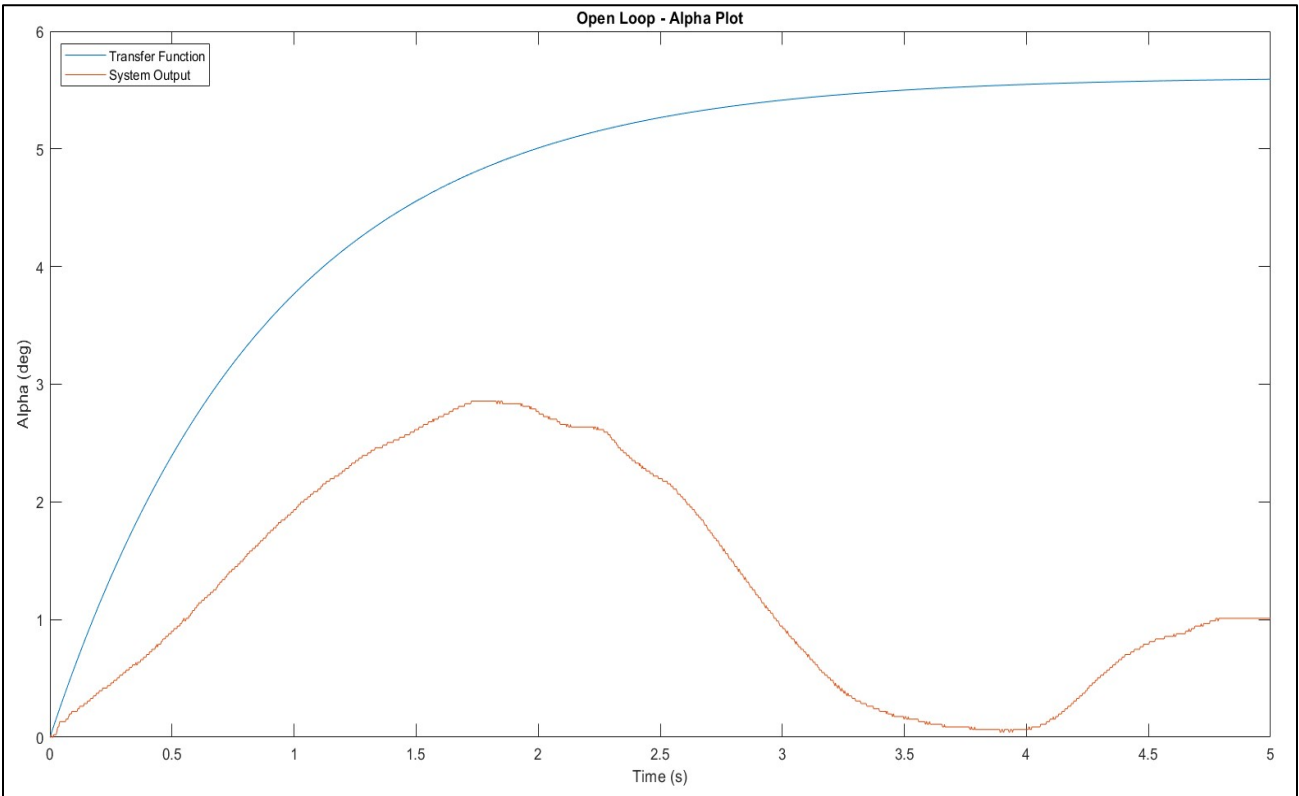


Figure 1: Open Loop Response – Alpha

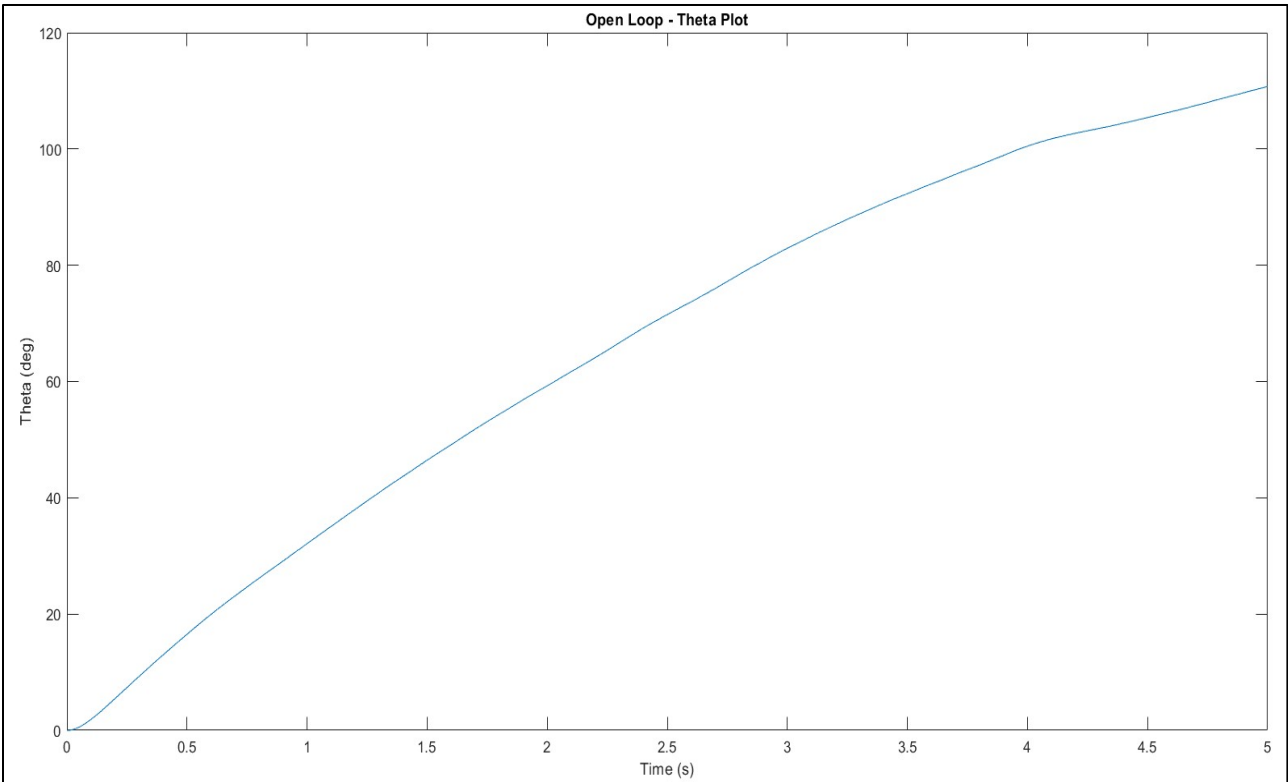


Figure 2: Open Loop Response – Theta

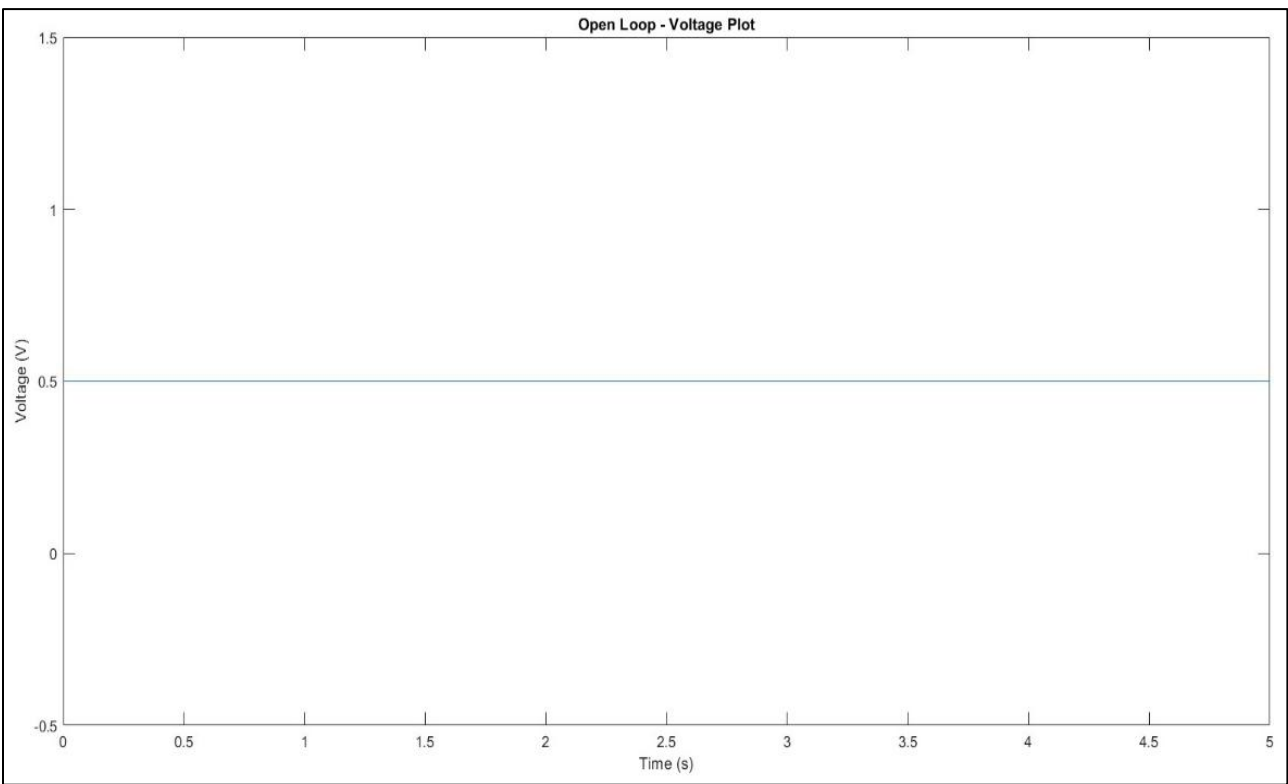


Figure 3: Open Loop Response – Voltage

D. Closed Loop Response PI Controller Plots

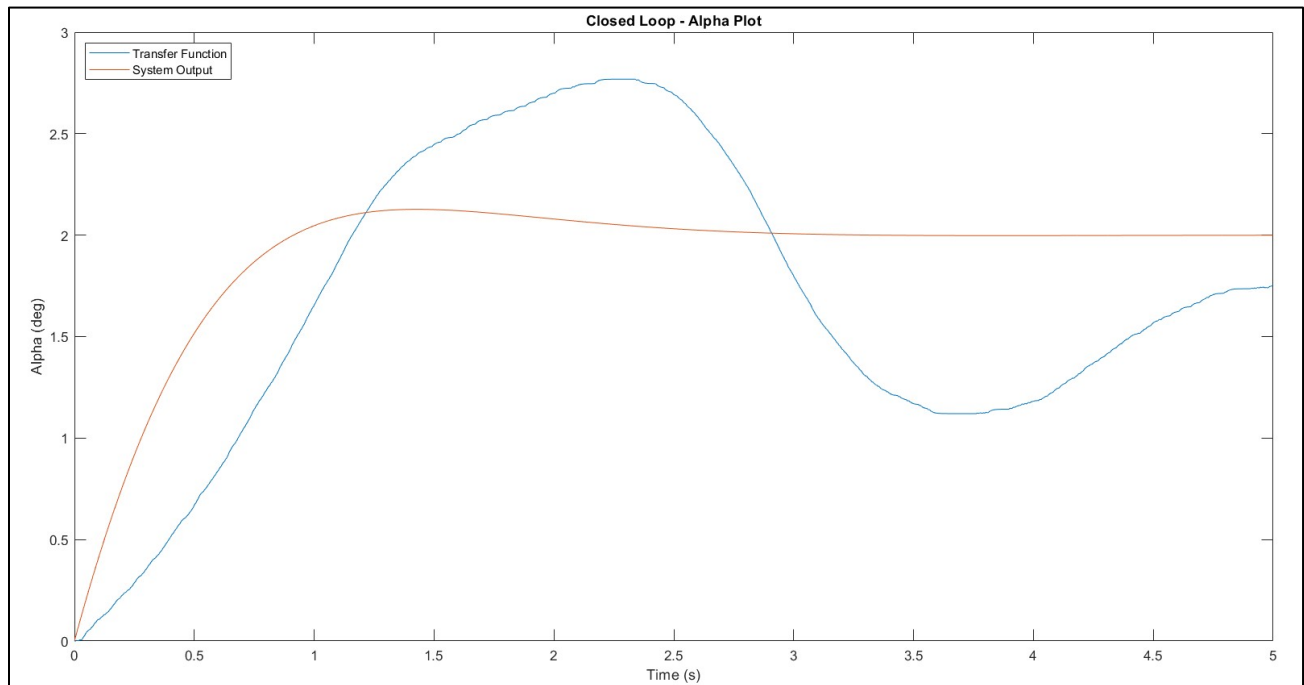


Figure 4: Closed Loop Response - Alpha

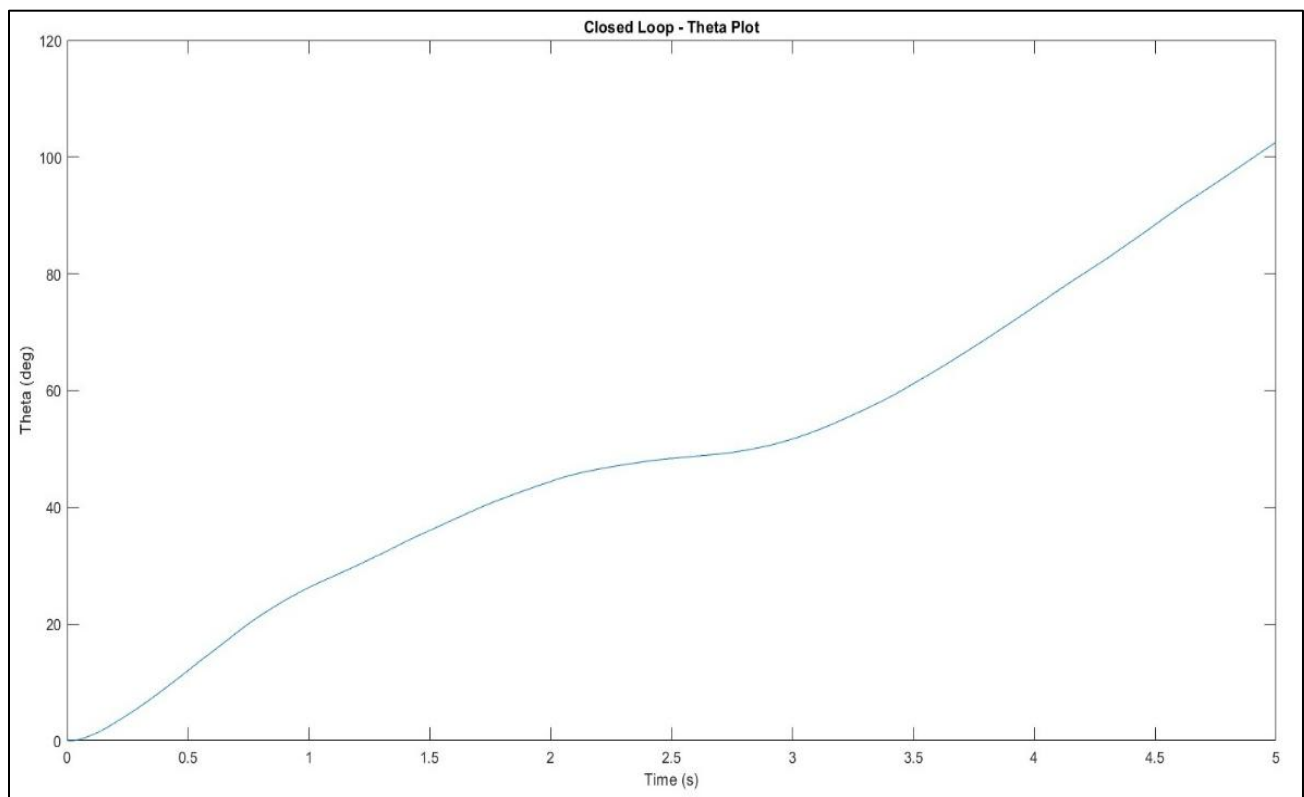


Figure 5: Closed Loop Response - Theta

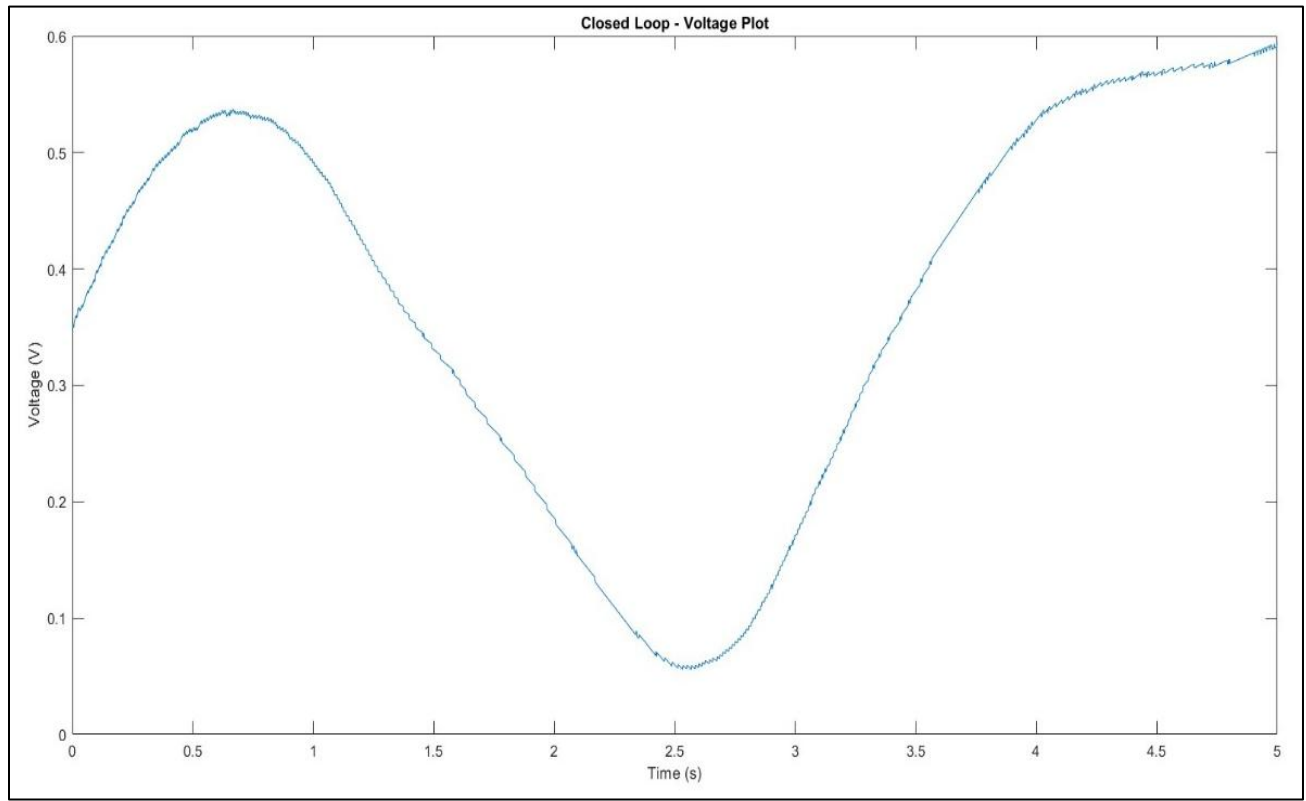


Figure 6: Closed Loop Response – Voltage

E. Closed Loop Response PI Controller with Disturbance Plots

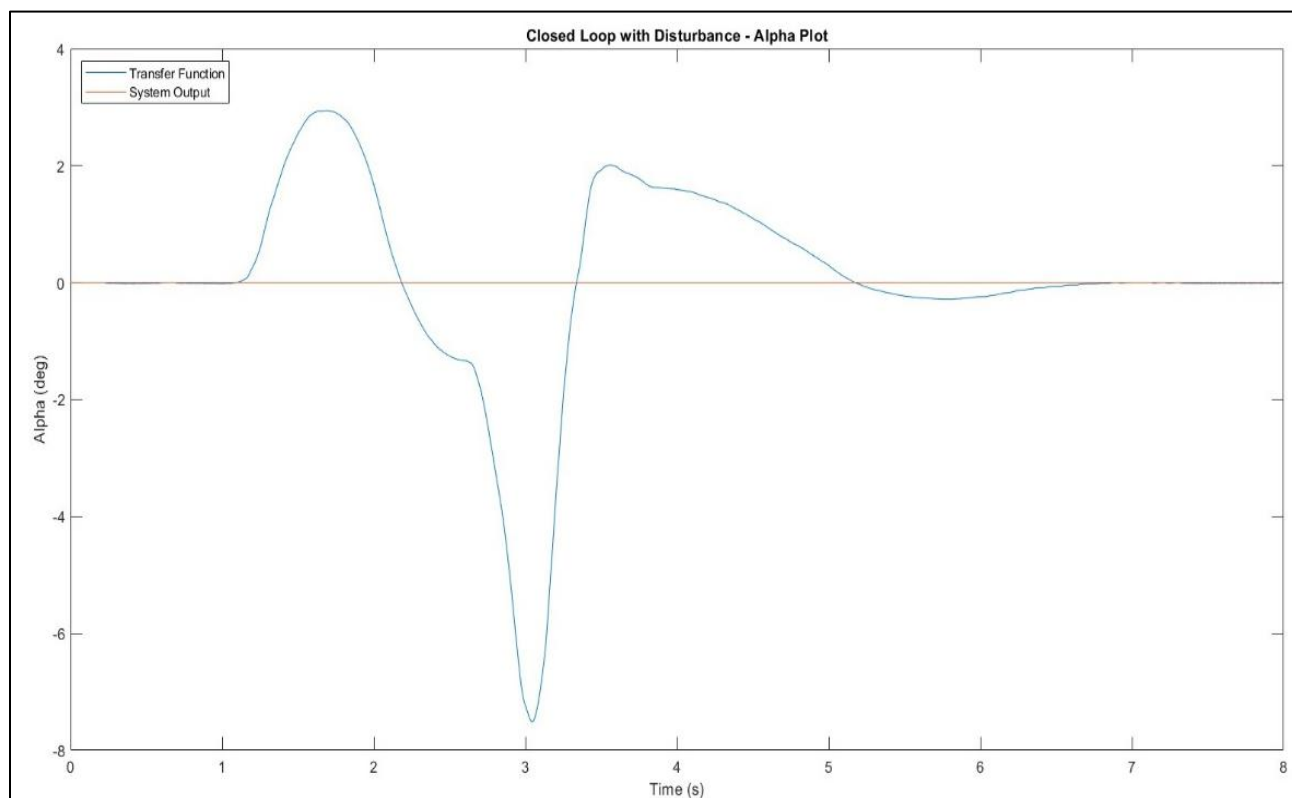


Figure 7: Closed Loop with Disturbance Response - Alpha

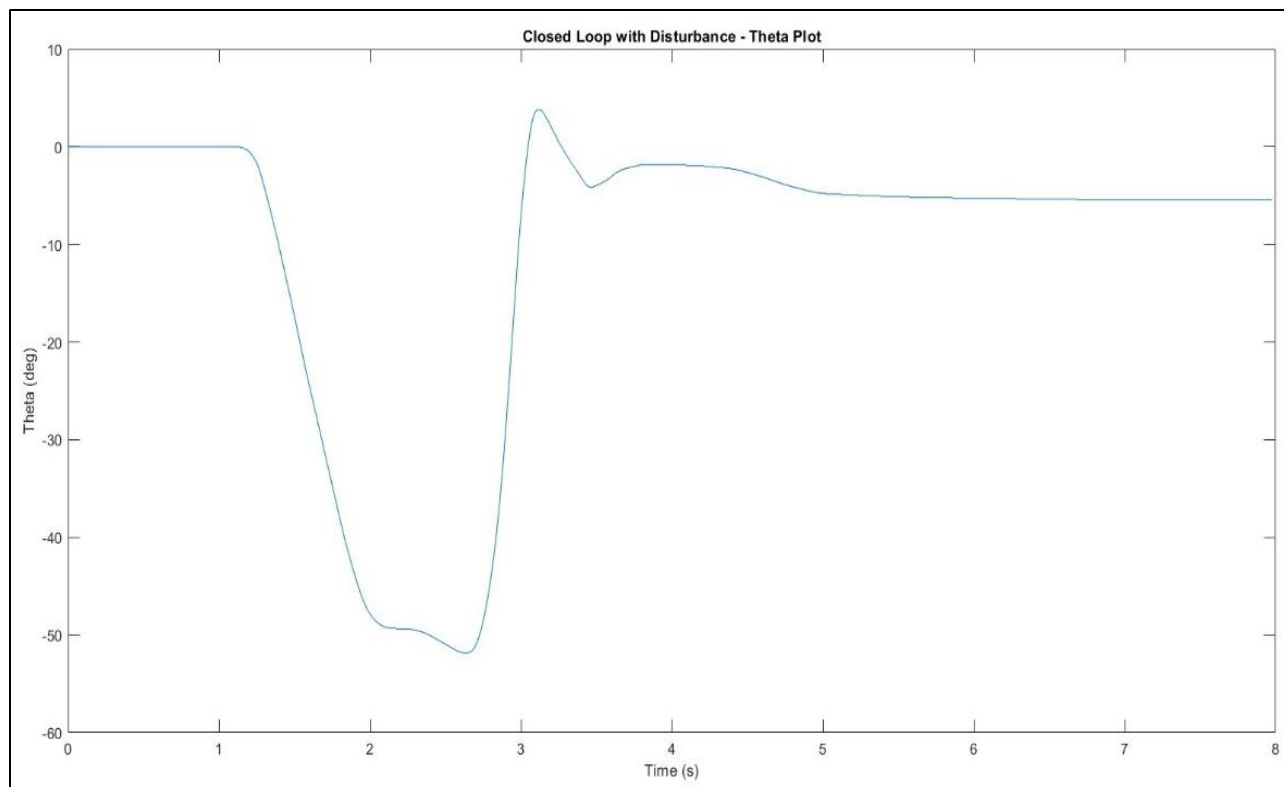


Figure 8: Closed Loop with Disturbance Response - Theta

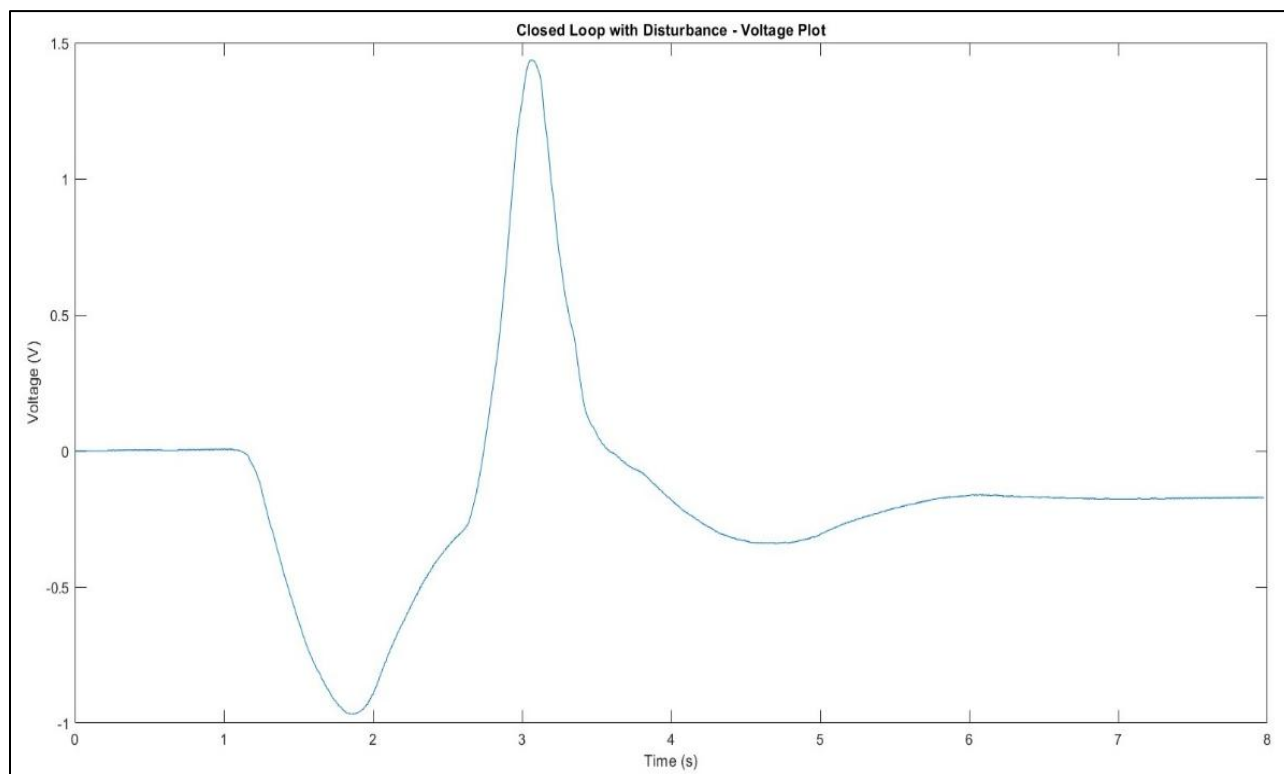


Figure 9: Closed Loop with Disturbance Response - Voltage

F. Open Loop Response PI Controller with Disturbance Plots

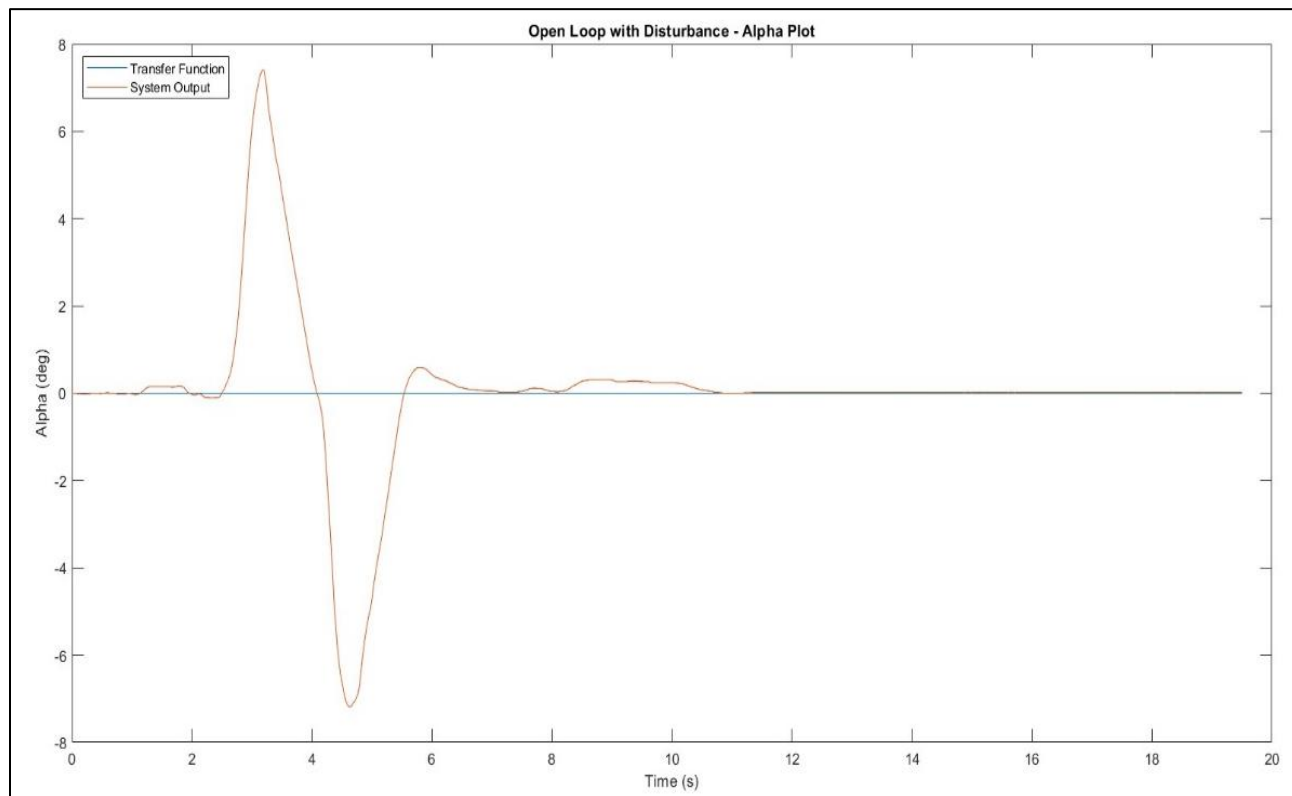


Figure 10: Open Loop with Disturbance Response - Alpha

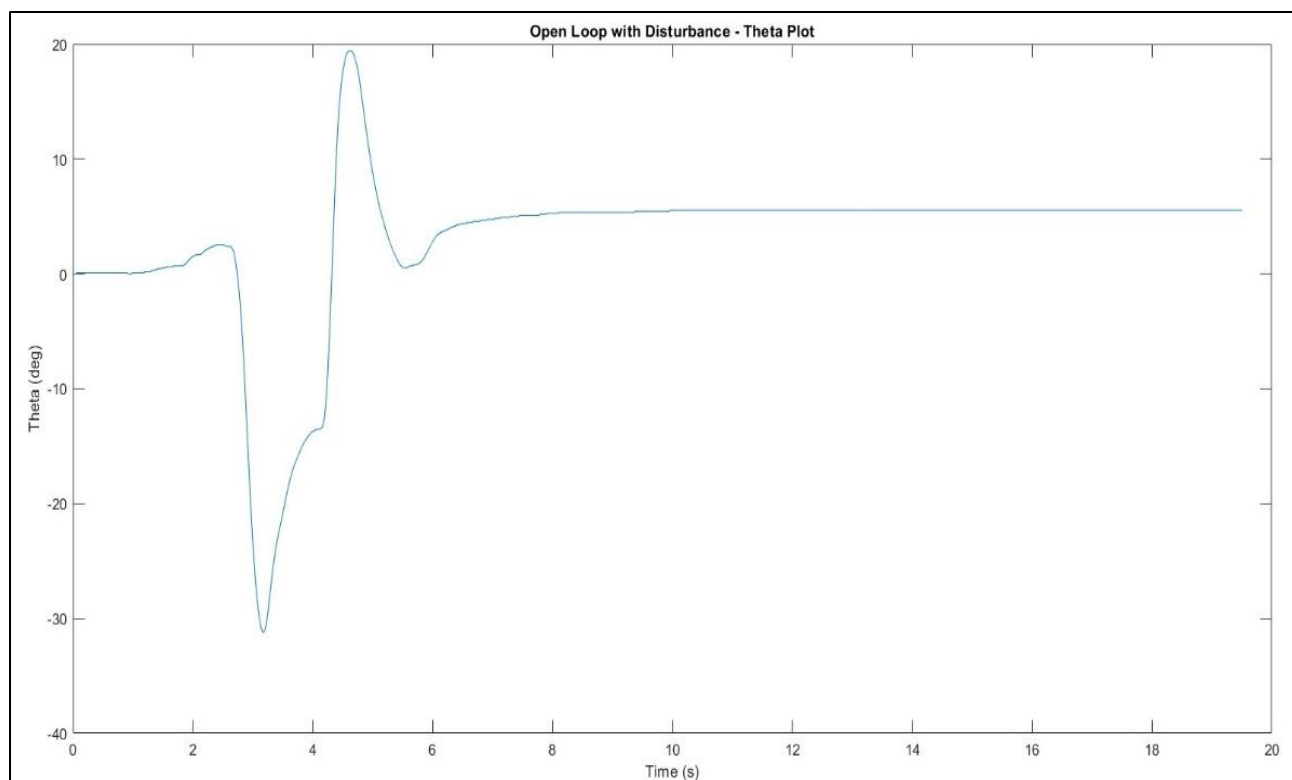


Figure 11: Open Loop with Disturbance Response - Theta

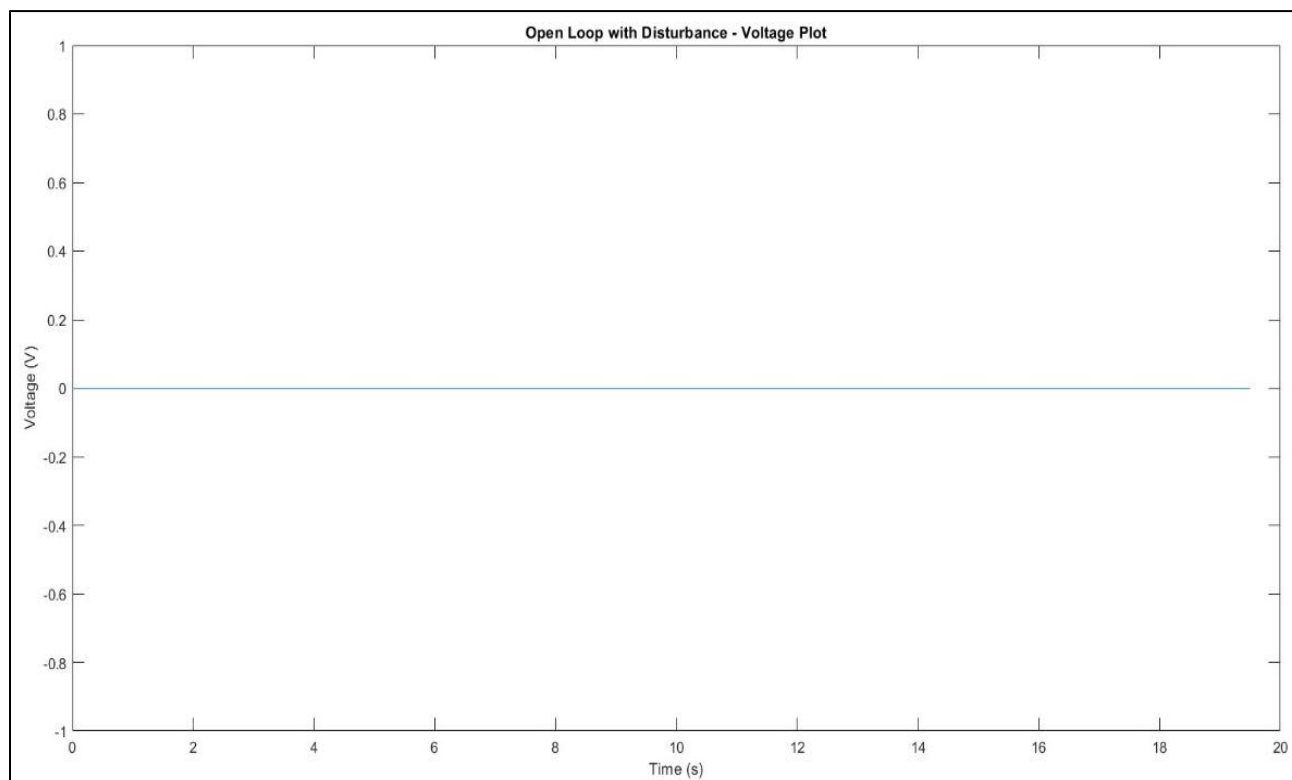


Figure 12: Open Loop with Disturbance Response - Voltage

Analysis

A. Evaluation of open-loop transfer function modeling

The accuracy of the open-loop transfer function model of the rotary gyroscope is evaluated by comparing the experimental results with the theoretical model, by referring to figure 1.

Theoretical Model

The open-loop transfer function relating the gyroscope deflection angle $\alpha(s)$ to the applied motor voltage $V_m(s)$ is given by

$$G_{OL}(s) = \frac{\alpha(s)}{V_m(s)} = \frac{1}{G_g} \frac{K}{\tau s + 1} \quad (1)$$

where

- $K = 1.53 \text{ rad/s/V}$ is the steady-state gain
- $\tau = 0.9 \text{ s}$ is the time constant
- $G_g = 5.20 \text{ 1/(rad} \cdot \text{s)}$ is the gyroscopic sensitivity gain.

This represents a first-order system, meaning the response should smoothly rise and approach a steady-state value.

Experimental Observations

From figures 1, 2, and 3, it can be determined that the input voltage V_m remains approximately constant at 0.5 V. The servo angle θ increases almost linearly with time, indicating that the base is rotating at a nearly constant angular velocity. This behavior is consistent with the servo model:

$$P(s) = \frac{K}{s(\tau s + 1)} \quad (2)$$

The deflection angle α shows a response that initially increases but then deviates significantly from the theoretical prediction. Instead of smoothly approaching a constant steady-state value, the measured response rises to a peak and then decreases, showing fluctuations.

Quantitative Comparison

From equation (1), at steady state, s equals to zero. Therefore, the open-loop transfer function becomes

$$\frac{\alpha_{ss}}{V_m} = \frac{K}{G_g} \quad (3)$$

$$\alpha_{ss} = \frac{K \cdot V_m}{G_g} \quad (4)$$

Using equation (4), theoretical deflection angle can be estimated:

$$\alpha_{ss} = \frac{1.53 \cdot 0.5}{5.20} = 0.147 \text{ rad} = 8.43 \text{ deg}$$

From figure 1, the experimental deflection angle α_{exp} is 2.8 deg at its peak.

Hence, the percentage error is

$$\% \text{ Error} = \left| \frac{\alpha_{theory} - \alpha_{exp}}{\alpha_{theory}} \right| \times 100 \quad (5)$$

$$\% \text{ Error} = \left| \frac{8.43 - 2.8}{8.43} \right| \times 100 = 66.8 \% \quad (6)$$

Conclusion

From both qualitative and quantitative analyses, the open-loop transfer function provides only a moderate to poor representation of the actual system behavior. While the model correctly predicts the general first-order nature of the response, it fails to capture the true dynamics of the gyroscope. The theoretical model predicts a smooth rise to a steady value, whereas the experimental response shows a peak followed by a decrease and oscillatory behavior.

This difference can be mainly due to sensor noise and hardware limitations, unmodeled gyroscopic coupling effects, which are significant in the physical system but might be ignored in the simplified model, and nonlinear friction and mechanical disturbances from the spinning disk. This highlights the need for feedback control to compensate for these unmodeled dynamics and improve system performance.

B. Effect of having the PI control on and tracking a $\alpha = 2 \text{ deg}$ rotation

The effect of enabling the PI controller is analyzed by comparing the measured gyroscope response with the theoretical closed-loop transfer function. In this part of the experiment, the desired reference was set to

$$\alpha_{ref} = 2^\circ \quad (7)$$

and the goal was to determine how well the controller forces the gyroscope deflection angle α to track this reference.

Theoretical Model

The closed-loop transfer function of the gyroscope with PI control is

$$G_{CL}(s) = \frac{\alpha(s)}{\alpha_{ref}(s)} = \frac{K(sk_p + k_i)}{G_g\tau s^2 + (Kk_p + G_g)s + Kk_i} \quad (8)$$

This is a second-order closed-loop system. Since the controller includes both proportional and integral action, the expected response is that the output should approach the reference value. The integral term is especially important because it accumulates the error over time and helps eliminate any steady-state error.

Experimental Observations

From figure 6, it is obtained that the control voltage V_m is continuously adjusted by the controller. Instead of remaining constant, the voltage changes over time depending on the tracking error. This confirms that the controller is actively responding to the gyroscope motion rather than simply applying a fixed input.

Figure 5 shows that the servo angle continues to change over time. This means the rotary servo is moving the base in order to generate the torque needed to maintain the desired gyroscope response.

Figure 4 shows the comparison between the transfer function output and the measured system output. The actual system output rises quickly and settles close to 2° , while the transfer function response deviates more and shows a larger variation.

Quantitative Comparison

From figure 13, the measured α output is approximately

$$\alpha_{exp} = 1.997^\circ \quad (9)$$

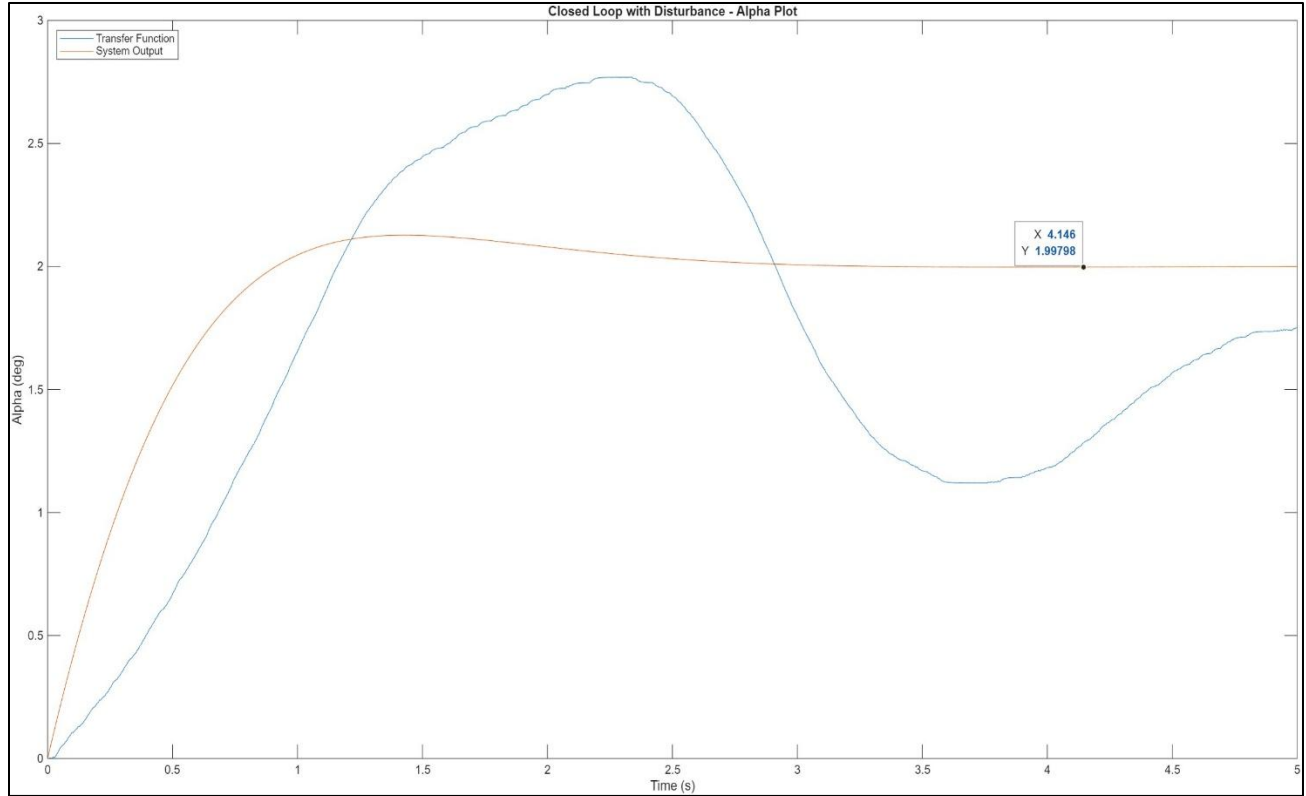


Figure 13: Closed Loop Response – Alpha

While the desired reference value:

$$\alpha_{ref} = 2.0^\circ \quad (10)$$

Thus, the steady-state error is:

$$e_{ss} = \alpha_{ref} - \alpha_{exp} \quad (11)$$

$$e_{ss} = 2.0 - 1.997 = 0.003^\circ \quad (12)$$

The steady-state error value is approximately zero. This indicates that the PI controller is able to remove the steady-state error and achieve accurate tracking of the desired gyroscope angle.

Conclusion

The PI controller improves the gyroscope response because it provides two separate effects, i.e., proportional and integral action. The proportional gain reacts immediately to the tracking error and generates a correcting motor voltage. The integral gain accumulates the error over time and forces the steady-state error toward zero. This is clearly reflected in the measured α response. The system output quickly

risers and settles near the commanded value of 2° , showing that the controller is successfully regulating the gyroscope. At the same time, the servo angle θ continues to rotate, which shows that the SRV02 motor is actively repositioning the base to maintain the gyroscope orientation.

However, although the closed-loop transfer function predicts the general response trend, there is still a difference between the theoretical and measured outputs. The measured system reaches the desired 2° value more directly, while the transfer function output shows a larger variation in its response and does not match the real system's response perfectly. This difference suggests that the theoretical model captures the main closed-loop behavior, but it still does not include all physical effects of the actual setup. The likely causes could be similar as in the previous experiment, such as hardware limitations, simplified model, unmodeled gyroscopic coupling effects, and mechanical disturbances from the spinning disk.

C. *Effect of having the PI control on and off*

In this part of the experiment, the effect of the PI controller is analyzed by comparing the gyroscope response when the controller is enabled and when it is disabled. The purpose is to determine how feedback control changes the system behavior when a disturbance is applied by manually rotating the base plate.

The desired reference for this part of the experiment is

$$\alpha_{ref} = 0^\circ \quad (13)$$

which means the controller should act so that the gyroscope deflection angle remains at zero even when the base plate is disturbed.

Theoretical Model

From the gyroscope kinematics, the total rotation of the gyro module relative to the base plate is

$$\eta = \gamma + \theta_l \quad (14)$$

Differentiating gives

$$\dot{\eta} = \dot{\gamma} + \dot{\theta}_l \quad (15)$$

Using the gyroscope sensitivity relation,

$$\omega_b = G_g \alpha \quad (16)$$

and since $\dot{\eta} = \omega_b$, we obtain

$$G_g \alpha = \dot{\gamma} + \dot{\theta}_l \quad (17)$$

This shows that the measured gyro deflection angle α is directly related to the disturbance motion and the servo motion.

The open-loop transfer function is

$$G_{OL}(s) = \frac{\alpha(s)}{V_m(s)} = \frac{1}{G_g} \frac{K}{\tau s + 1} \quad (18)$$

and the closed-loop transfer function with PI control is

$$G_{CL}(s) = \frac{\alpha(s)}{\alpha_{ref}(s)} = \frac{K(s k_p + k_i)}{G_g \tau s^2 + (K k_p + G_g) s + K k_i} \quad (19)$$

The PI controller law is

$$V_m(t) = k_p e(t) + k_i \int e(t) dt \quad (20)$$

where

$$e(t) = \alpha_{ref}(t) - \alpha(t) \quad (21)$$

Since $\alpha_{ref} = 0^\circ$,

$$e(t) = -\alpha(t) \quad (22)$$

therefore,

$$V_m(t) = -k_p \alpha(t) - k_i \int \alpha(t) dt \quad (23)$$

This means that when α deviates from zero due to a disturbance, the controller generates a motor voltage to oppose that deviation.

Case 1: PI Control ON

When the PI controller is enabled, the system actively responds to the disturbance. From figure 9, the motor voltage is not constant. It shows a large negative deviation followed by a strong positive peak and then gradually settles. This means the controller is actively generating corrective torque through the motor. The fact that $V_m(t)$ changes significantly confirm that the controller is reacting to the disturbance in real time.

Figure 8 shows that the servo angle changes rapidly when the disturbance is applied. The angle first moves strongly in one direction and then reverses before settling. This indicates that the SRV02 motor rotates the base to counter the imposed disturbance. The servo is therefore not passively following the motion and instead being driven by the controller so that the gyroscope heading is maintained.

From figure 9, the measured system output remained at zero throughout the experiment, while the theoretical transfer function response shows a much larger oscillatory behavior, but eventually settling at zero value. Thus, the measured steady-state error is zero:

$$e_{ss} = \alpha_{ref} - \alpha_{exp} \quad (24)$$

$$e_{ss} = 0 - 0 = 0^\circ \quad (25)$$

This is the main evidence that the PI controller is working. Even though the base is disturbed, the gyro deflection stays regulated near zero.

Case 2: PI Control OFF

When the PI controller is disabled, no active correction is applied by the motor.

In this case,

$$V_m(t) = 0 \quad (26)$$

which is confirmed directly by the open-loop voltage plot, where the applied motor voltage remains zero for the entire experiment. Figure 12 shows a flat line at zero, which means the motor is not generating any corrective action. Therefore, the system is left to respond only through its passive gyroscopic and mechanical dynamics.

From figure 11, the servo angle changes significantly when the base is rotated. Unlike the controlled case, this motion is not part of an active compensation mechanism.

The system simply reacts to the disturbance and settles at a different position. This indicates that without control, the servo cannot restore the original heading effectively.

From figure 10, the measured system output exhibits a large transient response. The deflection reaches approximately $+7.36^\circ$, then swings to about -7.185° , and only later returns toward zero, which is interpreted from figure 14.

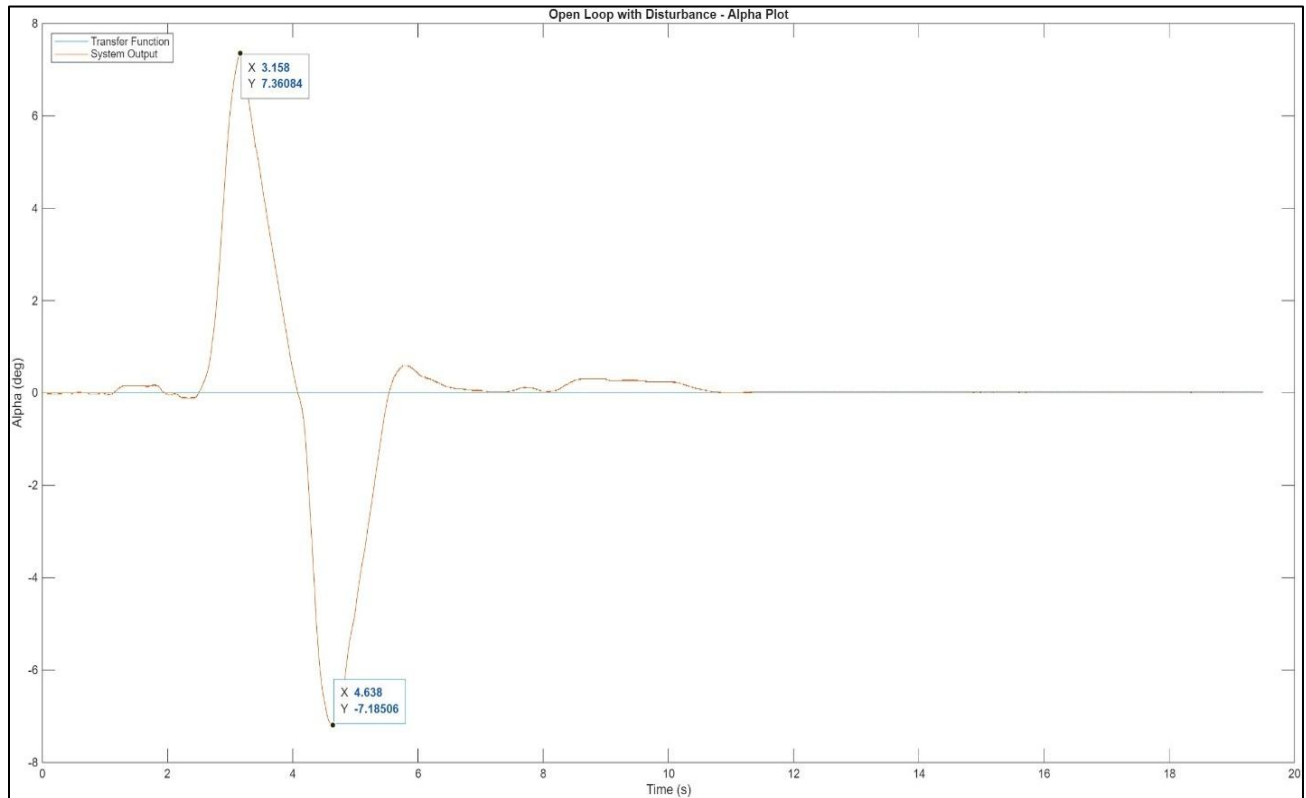


Figure 14: Open Loop with Disturbance Response - Alpha

This is a much larger disturbance response than the closed-loop case. Since the reference is still

$$\alpha_{ref} = 0^\circ \quad (27)$$

the error becomes

$$e(t) = 0 - \alpha(t) \quad (28)$$

and because no controller is acting, the system cannot immediately correct this deviation.

Although the response eventually returns close to zero due to passive system behavior, the deflection becomes very large during the disturbance. This means the gyroscope heading is not maintained well in the open-loop case.

Conclusion

The experimental results clearly show that the PI controller greatly improves disturbance rejection in the rotary gyroscope system. With PI control on, the motor voltage and servo angle are actively adjusted so that the gyro deflection angle stays near the desired value of 0° . This means the gyroscope is able to maintain its heading even when the base plate is rotated. In contrast, when PI control is off, the motor voltage remains zero and the gyroscope experiences large deflections before slowly returning toward equilibrium. Therefore, the PI controller is essential for converting the rotary gyroscope from a passive device into an actively stabilized system.

D. Stability Analysis Using the Routh-Hurwitz Criterion

To analyze the stability of the rotary gyroscope system, the Routh-Hurwitz criterion is applied to the characteristic equations of both the open-loop and closed-loop transfer functions.

Open-Loop Stability Analysis

The open-loop transfer function of the gyroscope is

$$G_{OL}(s) = \frac{\alpha(s)}{V_m(s)} = \frac{1}{G_g} \frac{K}{\tau s + 1} \quad (29)$$

The characteristic equation is obtained from the denominator:

$$\tau s + 1 = 0 \quad (30)$$

For a first-order system, the Routh-Hurwitz criterion requires that all coefficients of the characteristic equation be positive. In this case, τ (0.9 s) and 1 both greater than 0, hence positive. Therefore, the pole location is:

$$s = -\frac{1}{\tau} = -\frac{1}{0.9} = -1.11 \quad (31)$$

Since the pole lies in the left-half of the s-plane, the open-loop system is stable.

Closed-Loop Stability Analysis

With PI control, the closed-loop transfer function is

$$G_{CL}(s) = \frac{\alpha(s)}{\alpha_{ref}(s)} = \frac{K(sk_p + k_i)}{G_g\tau s^2 + (Kk_p + G_g)s + Kk_i} \quad (32)$$

The characteristic equation is therefore

$$A(s) = G_g\tau s^2 + (Kk_p + G_g)s + Kk_i = 0 \quad (33)$$

For a second-order polynomial written as $a_2s^2 + a_1s + a_0 = 0$, the Routh-Hurwitz criterion requires $a_2 > 0$, $a_1 > 0$, and $a_0 > 0$.

When comparing coefficients,

$$a_2 = G_g\tau \quad (34)$$

$$a_1 = Kk_p + G_g \quad (35)$$

$$a_0 = Kk_i \quad (36)$$

Using the nominal parameter values,

$$G_g = 5.20, \tau = 0.9, K = 1.53 \quad (37)$$

Thus, the stability conditions are

$$G_g\tau = 5.20(0.9) = 4.68 > 0 \quad (38)$$

which is satisfied.

The second condition gives

$$1.53 k_p + 5.20 > 0 \quad (39)$$

$$k_p > -3.40 \quad (40)$$

Since k_p is 19.79 from equation (xxi), this condition is satisfied.

The third condition gives

$$1.53 k_i > 0 \quad (41)$$

$$k_i > 0 \quad (42)$$

So, the closed-loop system is stable as long as the integral gain is positive and since k_i was calculated as 39.86 from equation xxiv, the system is stable.

Routh Table for the Closed-Loop System

For the characteristic equation

$$G_g \tau s^2 + (Kk_p + G_g)s + Kk_i = 0 \quad (43)$$

The Routh table is

s^2	$G_g \tau$	Kk_i
s^1	$Kk_p + G_g$	0
s^0	Kk_i	

Table 1: Routh-Hurwitz Criterion Table

For stability, all terms in the first column must be positive:

$$G_g \tau > 0 \quad (44)$$

$$Kk_p + G_g > 0 \quad (45)$$

$$Kk_i > 0 \quad (46)$$

Since these conditions are satisfied for positive k_p and k_i , the closed-loop system is stable.

Conclusion

Using the Routh-Hurwitz criterion, both the open-loop and closed-loop rotary gyroscope systems are stable. The open-loop system is stable because its single pole is located in the left-half plane. The closed-loop PI-controlled system is also stable because all coefficients of its characteristic equation are positive and all entries in the first column of the Routh table are positive. Therefore, the designed PI controller produces a stable closed-loop gyroscope system.

Conclusion

Effect of PI Control and Reason for Rotation about the z-axis

The experimental results clearly show that the introduction of PI control converts the system from an open-loop, oscillatory behavior into a stable and accurate tracking system for the commanded deflection $\alpha = 2^\circ$. In the absence of control, mechanical effects such as friction and damping prevent the system from maintaining a desired orientation. With PI control, this limitation is overcome through the combined action of proportional and integral terms.

The control law is given by

$$V_m(t) = k_p e(t) + k_i \int e(t) dt \quad (47)$$

where

$$e(t) = \alpha_{ref} - \alpha(t) \quad (48)$$

The proportional term $k_p e(t)$ provides an immediate corrective action proportional to the tracking error, improving response speed. The integral term $k_i \int e(t) dt$ accumulates the error over time and increases the control input until the steady-state error is eliminated. This explains why the system is able to reach and maintain $\alpha = 2^\circ$ with negligible steady-state error in the experimental results.

The observed rotation about the z-axis is explained by the physics of gyroscopic precession. The spinning disk possesses angular momentum H , and when a torque is applied to maintain the deflection angle, the system responds according to

$$T_g = H \cdot \dot{\theta} \quad (49)$$

This relationship shows that a non-zero base angular velocity $\dot{\theta}$ is required to generate the gyroscopic torque necessary to sustain a non-zero deflection α . Rearranging,

$$\dot{\theta} = \frac{T_g}{H} \quad (50)$$

indicates that maintaining a constant deflection requires continuous rotation about the z-axis.

Thus, when the PI controller enforces $\alpha = 2^\circ$, it does so by continuously adjusting the motor voltage to generate the required base rotation $\dot{\theta}$. This rotation is not an undesired effect but a necessary physical condition for equilibrium. If the base rotation were to stop ($\dot{\theta} = 0$), the gyroscopic torque would vanish, and the deflection angle would return to zero.

Relation to an Actual Gyroscope

The behavior observed in this experiment directly reflects how real gyroscope systems operate in applications such as aircraft, satellites, and inertial navigation systems. In these systems, the gyroscope acts as a stable platform, where the goal is to maintain a fixed orientation despite the motion of the base.

In this setup, the deflection angle α represents the platform orientation. When PI control is applied, the system maintains α at the desired value by continuously adjusting the motor voltage according to

$$V_m(t) = k_p e(t) + k_i \int e(t) dt \quad (51)$$

where $e(t) = \alpha_{ref} - \alpha(t)$. This feedback action allows the system to compensate for disturbances such as friction and base motion, resulting in negligible steady-state error.

Without control, the system behaves as an open-loop gyroscope, where the deflection angle is directly affected by base rotation and disturbances, leading to deviation and poor accuracy. This demonstrates that passive gyroscopic behavior alone is insufficient for precise control.

Thus, real gyroscope systems combine gyroscopic dynamics with closed-loop control to reject disturbances, eliminate deviation, and maintain accurate orientation, which is exactly what is observed in this experiment.

References

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Appendix A

MATLAB Code:

```
% Author: Anuranan Bharadwaj
% Lab 4
```

```

figure()
plot(data_alpha(:,1), data_alpha(:,2))
hold on
plot(data_alpha(:,1), data_alpha(:,3))
xlabel('Time (s)')
ylabel('Alpha (deg)')
title('Open Loop with Disturbance - Alpha Plot')
legend('Transfer Function', 'System Output', Location='northwest')

%%
plot(data_theta(:,1), data_theta(:,2))
xlabel('Time (s)')
ylabel('Theta (deg)')
title('Open Loop with Disturbance - Theta Plot')

%%
plot(data_vm(:,1), data_vm(:,2))
xlabel('Time (s)')
ylabel('Voltage (V)')
title('Open Loop with Disturbance - Voltage Plot')

%%
figure()
plot(data_alpha_PI(:,1), data_alpha_PI(:,2))
hold on
plot(data_alpha_PI(:,1), data_alpha_PI(:,3))
xlabel('Time (s)')
ylabel('Alpha (deg)')
title('Closed Loop with Disturbance - Alpha Plot')
legend('Transfer Function', 'System Output', Location='northwest')

```